

RS002 Week 13 — English Worksheet

Topic: Random Battles — `import random` · **Course:** Pokédex App · **Time:** about 45 minutes

Words to know: **import, random, choice, score, if / elif / else**

This week your dex becomes a game. In your live lesson you used `import random` to pick wild Pokémon out of nowhere, then compared stats with `if` / `elif` / `else` to decide a winner. This worksheet is about reading that code, thinking about what it does, and explaining it in your own words.

1 · Recall

From memory:

Write a function that takes a dictionary `p` and prints its name and attack.

2 · Predict 🧠

Read each block. Write what you think will happen.

```
import random

wild = ["꼬렛", "구구", "캐터피"]
print(random.choice(wild))
print(random.choice(wild))
```

Could the two printed names be different? Could they be the same? Why?


```
a = {"name": "피카츄", "attack": 55}
b = {"name": "이상해씨", "attack": 49}

if a["attack"] > b["attack"]:
    print(f"{a['name']} 승리!")
else:
    print(f"{b['name']} 승리!")
```

Which name is printed? Why?


```
import random

dex = [{"name": "피카츄"}, {"name": "파이리"}]
mine = random.choice(dex)
print(mine["name"])
```

`random.choice` here picks a whole dictionary. What kind of thing is `mine` , and what prints?

3 · Spot the Bug 🐛

Read what each block is **supposed** to do, find the problem, then explain the fix on paper.

Bug A — This should pick a random Pokémon. Right now `random` was never imported, so it crashes.

```
dex = ["피카츄", "이상해씨", "파이리"]  
print(random.choice(dex))
```

Hint: you must bring the tool in before using it.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — A draw (equal attack) should print `무승부`. Right now equal stats wrongly count as a loss.

```
a = {"name": "피카츄", "attack": 50}  
b = {"name": "파이리", "attack": 50}  
  
if a["attack"] > b["attack"]:  
    print(f"{a['name']} 승리!")  
else:  
    print(f"{b['name']} 승리!")
```

Hint: add an `elif` for `<` and an `else` for the tie.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This should compare two Pokémon's stats. Right now it compares a Pokémon to its own name string and crashes.

```
import random
dex = [{"name": "피카츄", "attack": 55}, {"name": "파이리", "attack": 52}]

mine = random.choice(dex)
enemy = random.choice(dex)

if mine["attack"] > enemy:
    print("승리!")
```

Hint: compare `enemy["attack"]`, not `enemy` itself.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Explain the Code 📖

Read this working stat battle. You do not need to run it — read it and answer.

```
import random

pokedex = [
    {"name": "피카츄", "hp": 35, "attack": 55},
    {"name": "이상해씨", "hp": 45, "attack": 49},
    {"name": "파이리", "hp": 39, "attack": 52},
]

my_pokemon = random.choice(pokedex)
enemy = random.choice(pokedex)

print(f"\n 🍒 {my_pokemon['name']} vs {enemy['name']}")
print(f"   체력: {my_pokemon['hp']} vs {enemy['hp']}")

my_score = my_pokemon["hp"] + my_pokemon["attack"]
enemy_score = enemy["hp"] + enemy["attack"]

if my_score > enemy_score:
    print(f"\n 🏆 {my_pokemon['name']}의 승리!")
elif my_score < enemy_score:
    print(f"\n 😞 {enemy['name']}에게 졌습니다...")
else:
    print(f"\n 🕊️ 무승부!")
```

What does `random.choice(pokedex)` give back — a name, a number, or a whole dictionary?

How is `my_score` calculated? Which two stats are added together?

When does the program reach the `elif` line instead of the first `if` ?

When does the `else` line run, and what does it print?

If you ran this program twice, why might you see a different match-up each time?

5 · Explain Your Lesson Code 🎥

In your live lesson you wrote a **Battle!** feature. Now explain *that* code — the one you wrote — in a short phone video. You may show it running. Teach it like the viewer is new, and use these key words: **import, random, choice, score, if / elif / else**.

1. Run your battle a few times and point out the different match-ups.
2. Read your `random.choice` line out loud and say what it picks.
3. Explain how your code decides the winner from the stats.
4. Say why the result changes each run.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.